- The Hyperledger framework focuses on computation logic and currently does not support cryptocurrency. The Hyperledger framework has many implementations, including Iroha, Intel's Sawtooth, Fabric, Indy, Burrow, and IBM's Fabric (version 2). Currently, these platforms are purely business-logic-based. No cryptocurrency is involved.

Thus in figure 12.8, you can see the two ends of the spectrum of blockchains, from real cryptocurrency enablers to pure business logic enablers. When you are designing blockchain-based solutions, you must consider cryptocurrency to be an integral part. Blockchain may not be only for sending and receiving value. You can create incentive models involving participants, fees for various activities, creative solutions for planetary-level problems (such as MPC discussed in chapter 7), and new economic models centered on a working digital currency. The cryptocurrency incurred is the cost of trust for the decentralized blockchain-based application.

12.11 Accessing off-chain data (Oracles)

Have you wondered how a smart contract can access external data? A smart contract operates in a sandbox. It cannot call an outside function or link to an external resource. Why can't a smart contract access outside sources? Depending on the source called, external data access from a smart contract may affect the global consistency of the blockchain. The results of any operation on the blockchain have to be deterministic. These issues limit the applicability of smart contracts in many real-world applications that may involve obtaining facts, data, and